Carbon-Aware Smart Building Energy Controller

Learning *when* to cool an office so the grid burns less carbon

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Why: when a building consumes matters as much as how much

- Buildings use **~30% of global final energy**; in Singapore, air-conditioning dominates commercial electricity
- Grid **carbon intensity changes hour to hour** with wind and solar output
- Buildings store "coolth" in their thermal mass → they *can* shift load in time
- But thermal inertia means an action now affects comfort **hours later** → a sequential decision problem

Question: can an RL controller cut HVAC carbon beyond a well-tuned carbon-aware rule, without hurting comfort?

What we built

1. **Simulator** of an office zone (1R1C thermal model) with a year of weather, occupancy, price and carbon data
2. **Formal MDP** with a multi-objective reward (carbon, cost, comfort)
3. **PPO and DQN** vs three rule-based controllers and an **exact DP upper bound**
4. **Analysis**: 5 seeds, held-out months, 95% CIs, Pareto sweep, state ablations, policy traces
5. **Responsible-AI** reading of what the agent actually learned

Simulator: a 1R1C office zone

$C \cdot dT/dt = (T_{out} - T_{in})/R + Q_{int} + Q_{sol} - COP \cdot P$

- $R = 0.4$ K/kW, $C = 15$ kWh/K → **$\tau = 6$ h**
- 14 kW HVAC, COP falls on hot afternoons
- Exact discretisation, 15-min steps
- **Validated:** step response hits 63% at τ; energy ≈ office ACMV range; DP oracle dominates every baseline (10 unit tests)

Series	Source
Weather	Singapore-like, 25–33 °C, rainy days
Occupancy	Weekday 08–18 + jitter, holidays
Price	USEP-like, SGD/kWh
Carbon intensity	**UK-grid-like** (SG grid ~95% gas → nearly flat CI)

Train Jan–Sep · **test Oct–Dec (92 unseen days)**

The MDP

- **State (20 features):** T_in, T_out, occupancy sensor, price, CI, time of day, weekday, forecasts of T_out / price / CI at +1, 2, 4, 8 h
- **Action:** HVAC power ∈ {0, 25, 50, 75, 100}% — the same space for PPO, DQN *and* exact DP
- **Reward:** $r_t = -[\lambda_C \cdot CI_t \cdot E_t + \lambda_\$ \cdot p_t \cdot E_t + \lambda_K \cdot \text{discomfort}_t]$, discomfort = °C·h outside **23–26 °C, only when occupied**
- **$\gamma = 0.99$** at 15 min ≈ 25 h effective horizon, long enough to credit 5 am pre-cooling for 2 pm comfort

Key observation: carbon and cost are both ∝ energy, so carbon-awareness = **shifting when you consume**. Price and CI are nearly uncorrelated here ($r = -0.07$): the cheapest hour is not the cleanest.

Controllers, and a fair fight

Controller	What it does
Fixed setpoint	Holds 24 °C around the clock
Setback	Cools during working hours only; **setpoint and start time tuned**
Carbon heuristic	Cools deeper when CI is low, drifts warmer when high; **4 parameters tuned**
Carbon-proportional	Setpoint rises smoothly with CI; **gain and base tuned**
DQN / **PPO**	PPO: clipped surrogate, GAE, return normalisation; **12 settings × 2 seeds tuned**
Oracle (DP)	Backward induction with perfect foresight, a true upper bound

Same rules for everyone: tuned on training days only (972 rule settings), scored with the same reward PPO maximises, then tested **once** on 92 unseen days. 5 seeds per learner, paired 95% bootstrap CIs.

Result: tuning matters most; PPO adds a little more

Controller	kg CO_2/day	Viol.*
Fixed setpoint	18.96	0%
Setback, 24 °C (untuned)	11.10	0%
Best tuned rule (25.9 °C)	8.52	0.7%
DQN	8.52	7.6%
PPO (tuned)	**8.21**	5.2%
Oracle (DP)	7.83	3.7%

*occupied 15-min steps outside 23–26 °C

Tuning one setpoint: **–23%** carbon

PPO vs the best tuned rule:

–3.7%

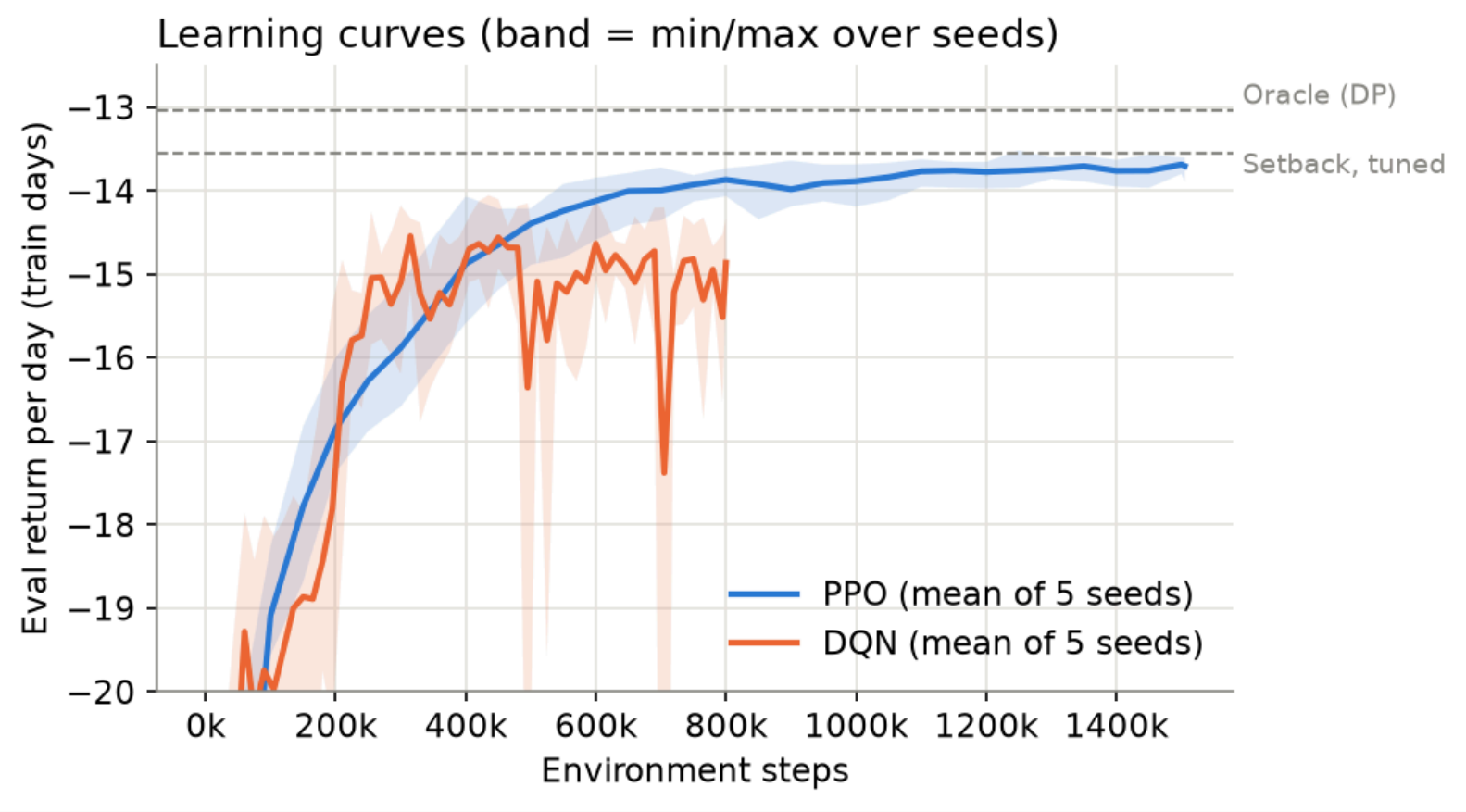
carbon, 95% CI [–4.0, –3.2]%

reward +1.1% (p = 0.03), cost –4.1%

19% of the gap to the oracle closed

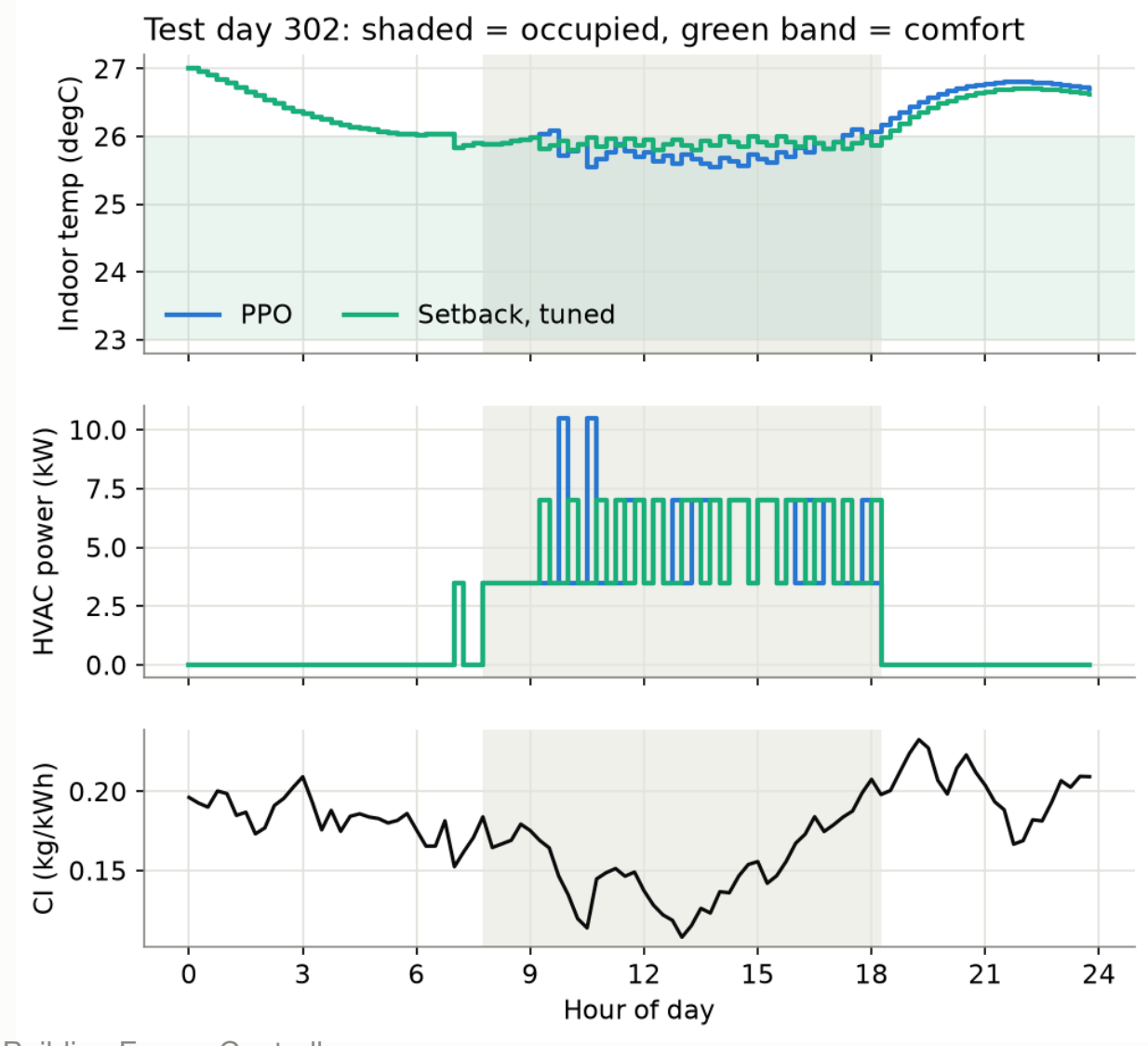
Price: more frequent but tiny violations (0.08 °C·h/day)

Learning: PPO is stable, DQN is noisy



PPO plateaus by ~700k steps; its 5 seeds land within 1.3% on test carbon. It ends **level with the tuned rule** on these 16 days and slightly ahead on 60 validation and 92 test days: a small edge.

What did PPO learn? Ride the band edge, cool a little in carbon dips

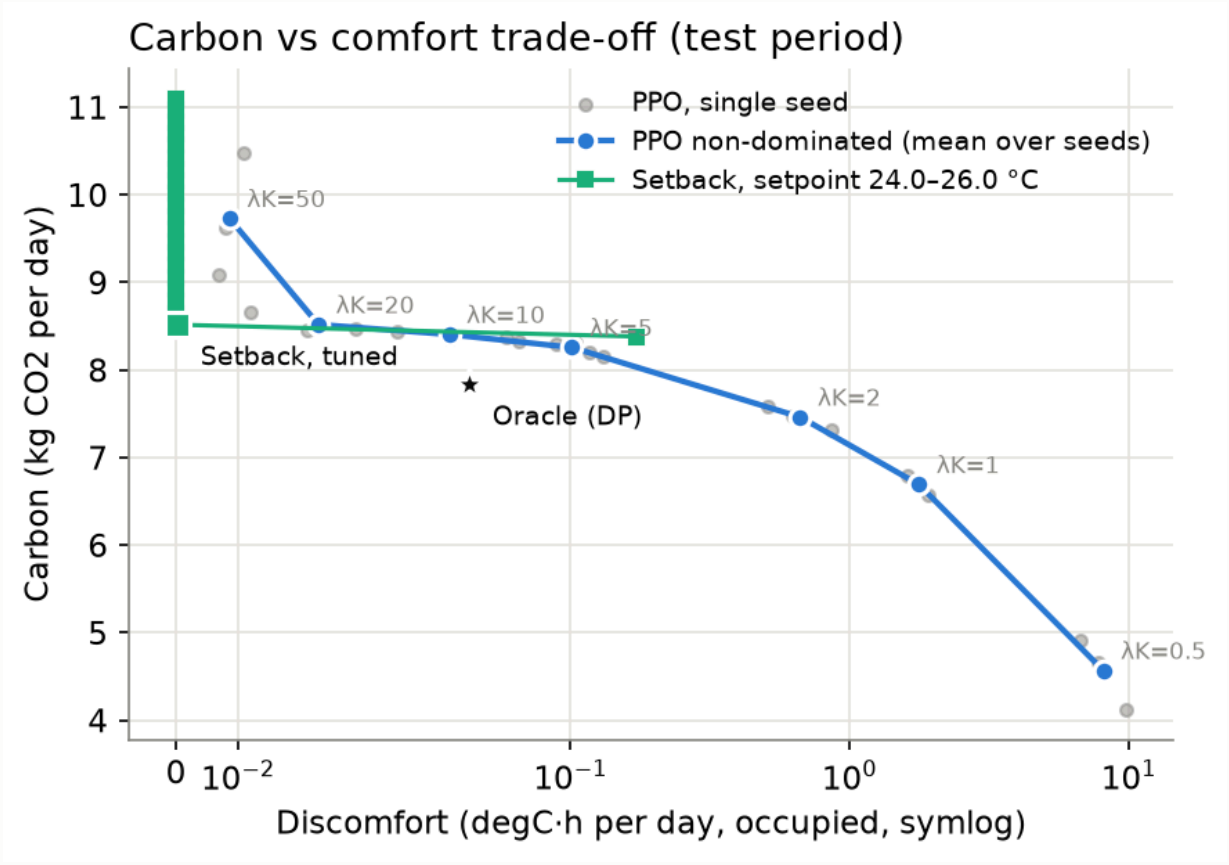


Where is the remaining headroom? Not in the carbon signal

Oracle variant (test)	Reward	kg CO_2
Full foresight	−13.07	7.83
No carbon foresight	−13.15	8.01
No occupancy foresight	**−14.08**	7.92
Best tuned rule	*−13.88*	*8.52*
PPO (tuned)	*−13.72*	*8.21*

- Knowing future CI is worth **0.08** reward/day
- Knowing when people arrive and leave is worth **~1.0**
- Tuning drove every carbon-reactive rule to **ignore CI**
- Removing CI from PPO's state costs only +1.3% carbon
- One room with τ = 6 h is a **leaky battery**: little to shift

At matched comfort, it's close to a tie



Inside the comfort band, the PPO front and the setpoint sweep overlap. PPO's extra savings come from **accepting more discomfort**.

Responsible AI

- **Who sets λ?** The weights decide whose comfort is traded for carbon. The Pareto front makes that trade visible.
- **Specification gaming:** PPO sits at the edge of the band. Comfort is charged only when occupied, so a faulty sensor invites exploitation; we **score comfort on true occupancy**.
- **Comfort is a distribution: 53%** of PPO's violations fall at 17:00–18:30. **Late leavers** bear them; the mean hides it.
- **Honest baselines:** an untuned baseline overstated PPO's gain from 3.7% to 11%. Comparing against weak rules is a quiet way to mislead.
- **Transparency:** a tuned thermostat gets most of the benefit and is fully explainable. That is a legitimate deployment choice.

Limitations and what is next

- Single zone, simplified thermal model, **synthetic profiles** (UK CI substituted for Singapore)
- Small RL margin; DQN, sweep and ablations used default PPO settings

In progress for the report

- **Occupancy prediction**: where the oracle's headroom actually is
- Robustness: shifted CI (reads signal vs memorises clock), noisy occupancy, tariff regimes
- **Constrained PPO** (Lagrangian): comfort as a constraint, not a weight
- **Battery storage**: a real lever for carbon shifting

Takeaways

1. **Tune the baseline first:** one setpoint gives –23% carbon; skipping this inflated our RL gain about 3×
2. Tuned PPO still wins, **–3.7% carbon and +1.1% reward**, with a little genuine load-shifting
3. The carbon signal is worth little in one room; **occupancy foresight** is the real headroom
4. An oracle plus fair tuning is what reveals all of this

Thank you · Questions?